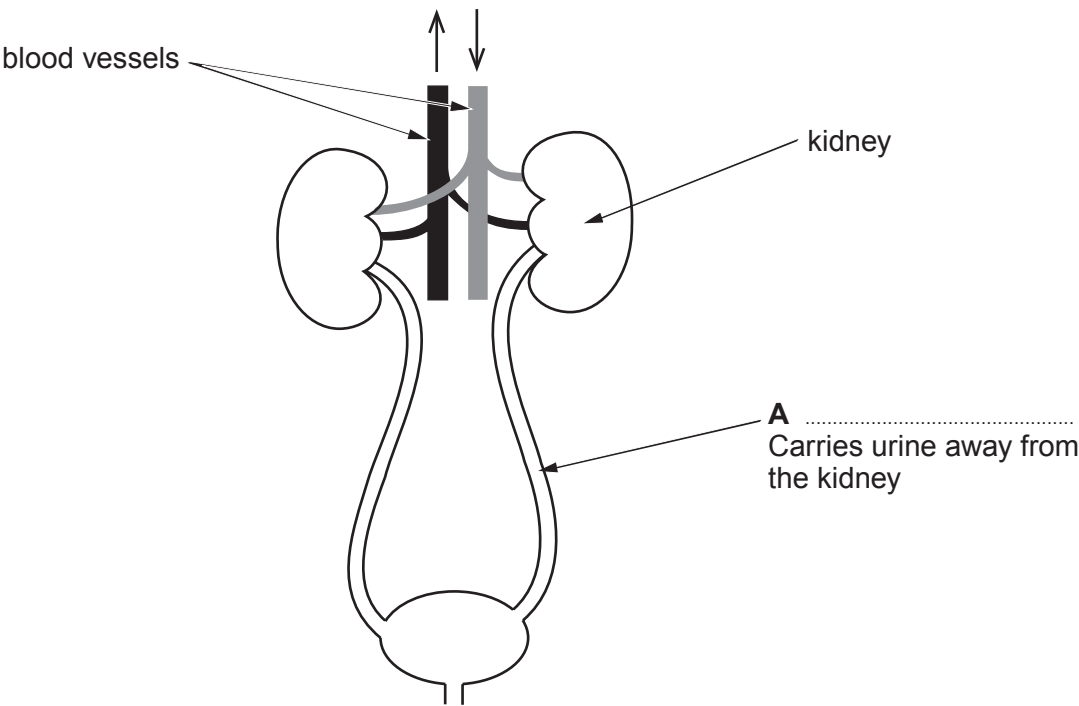


5. The diagram shows the human excretory system.



- (a) **Complete** label **A** on the diagram. [1]
- (b) A doctor investigated the concentrations of some substances present in the blood entering and leaving a patient's kidneys.

Substance	Concentration in blood (a.u.)	
	blood entering kidney	blood leaving kidney
glucose	168	168
salt (sodium)	35	33
protein	180	150
urea	314	11

- (i) From the table, state the waste substance which would be present in the patient's urine at the highest concentration. [1]

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(ii) The doctor thought that this patient’s urine would contain protein but no glucose. What is the evidence in the table to support this? [2]

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(iii) The urine was tested to find out if the doctor was correct.

I. Protein test

State the name of the chemical reagent added to the urine to test for the presence of protein and the colour which would indicate a positive result. [2]

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.....

II. Glucose test

When a sample of urine was tested for glucose a chemical reagent was added and the mixture was heated strongly. The colour remained blue showing the doctor to be correct.

Give the name of this chemical reagent and explain how the result would be different if the doctor was incorrect. [2]

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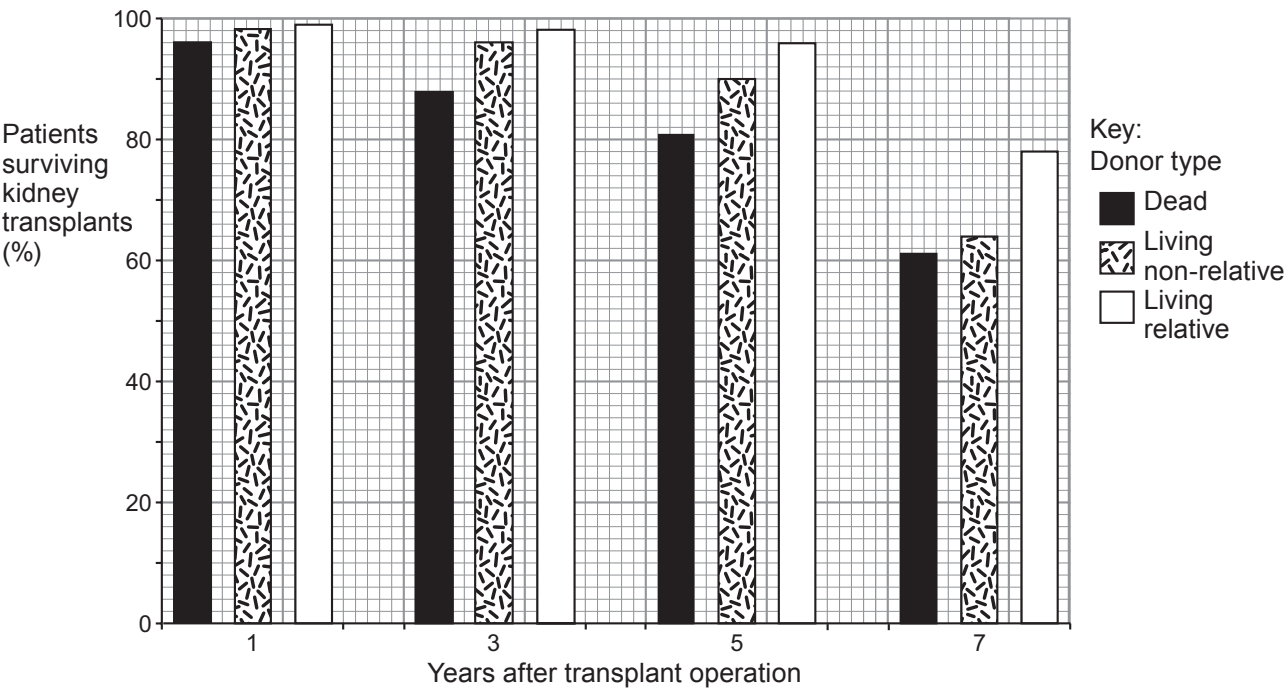
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(c) In the case of serious kidney disease a transplant operation can be carried out if a suitable kidney is available from a donor.

The bar chart shows the results of kidney transplant operations with donors of different types. All the recipients and donors were aged 30 – 50 years.



From the bar chart state **three** conclusions that can be drawn to compare the survival of patients after kidney transplant operations with different types of donors. [3]

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(d) Insufficient donor kidneys are available. What other form of treatment can be offered to people with kidney disease? [1]

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